Example 6.19 A tower stands vertically on the ground. From a point on the ground, which is 48 m away from the foot of the tower, the angle of elevation of the top of the tower is $30°$. Find the height of the tower.

Solution Let PQ be the height of the tower.

Take $PQ = h$ and QR is the distance between the tower and the point R. In the right angled ΔPQR, $\angle PRQ = 30°$

$$\tan\theta = \frac{PQ}{QR}$$

$$\tan 30° = \frac{h}{48} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{48} \Rightarrow h = 16\sqrt{3}$$

Therefore, the height of the tower is $16\sqrt{3}$ m

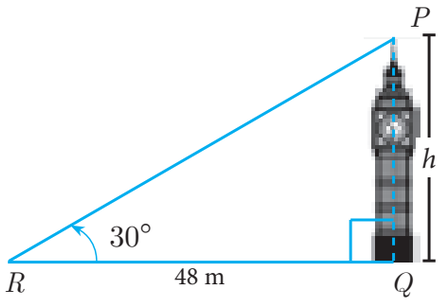


Fig. 6.13

Example 6.20 A kite is flying at a height of 75 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is $60°$. Find the length of the string, assuming that there is no slack in the string.

Solution Let AB be the height of the kite above the ground. Then, $AB = 75$.

Let AC be the length of the string.

In the right angled ΔABC, $\angle ACB = 60°$

$$\sin\theta = \frac{AB}{AC}$$

$$\sin 60° = \frac{75}{AC}$$

$$\Rightarrow \quad \frac{\sqrt{3}}{2} = \frac{75}{AC} \Rightarrow AC = \frac{150}{\sqrt{3}} = 50\sqrt{3}$$

Hence, the length of the string is $50\sqrt{3}$ m.

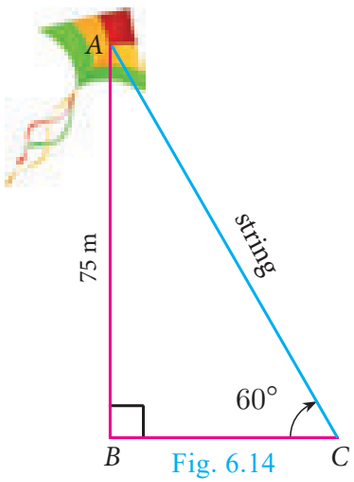


Fig. 6.14

Example 6.21 Two ships are sailing in the sea on either sides of a lighthouse. The angle of elevation of the top of the lighthouse as observed from the ships are $30°$ and $45°$ respectively. If the lighthouse is 200 m high, find the distance between the two ships. ($\sqrt{3} = 1.732$)

Solution Let AB be the lighthouse. Let C and D be the positions of the two ships.

Then, $AB = 200$ m.

$\angle ACB = 30°$, $\angle ADB = 45°$

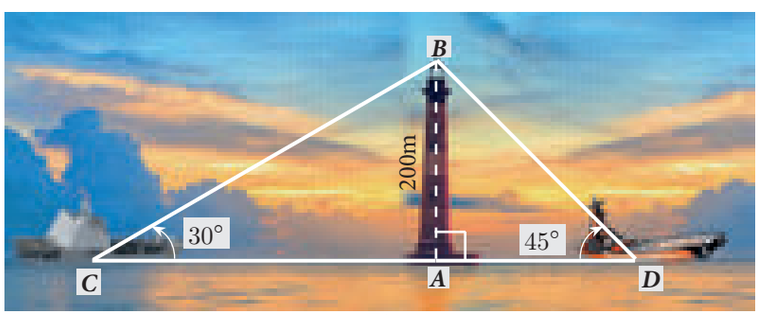


Fig. 6.15

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